

Project Integration Management



The Key to Overall Project Success: Good Project Integration Management

- ▶ Project managers must coordinate all of the other knowledge areas throughout a project's life cycle
- ▶ Many new project managers have trouble looking at the “big picture” and want to focus on too many details
- ▶ Project integration management is *not* the same thing as software integration

Project Integration Management Processes, Part 1

- ▶ **Developing the project charter** involves working with stakeholders to create the document that formally authorizes a project—the charter.
- ▶ **Developing the project management plan** involves coordinating all planning efforts to create a consistent, coherent document—the project management plan.
- ▶ **Directing and managing project work** involves carrying out the project management plan by performing the activities included in it.

Project Integration Management Processes, Part 2

- ▶ **Monitoring and controlling project** work involves overseeing activities to meet the performance objectives of the project
- ▶ **Performing integrated change** control involves identifying, evaluating, and managing changes throughout the project life cycle.
- ▶ **Closing the project or phase** involves finalizing all activities to formally close the project or phase.

Project Integration Management Summary

Initiating

Process: **Develop project charter**

Output: Project charter

Planning

Process: **Develop project management plan**

Output: Project management plan

Executing

Process: **Direct and manage project work**

Outputs: Deliverables, work performance data, change requests, project management plan updates, project documents updates

Monitoring and Controlling

Process: **Monitor and control project work**

Outputs: Change requests, project management plan updates, project documents updates

Process: **Perform integrated change control**

Outputs: Approved change requests, change log, project management plan updates, project documents updates

Closing

Process: **Close project or phase**

Outputs: Final product, service, or result transition; organizational process assets updates

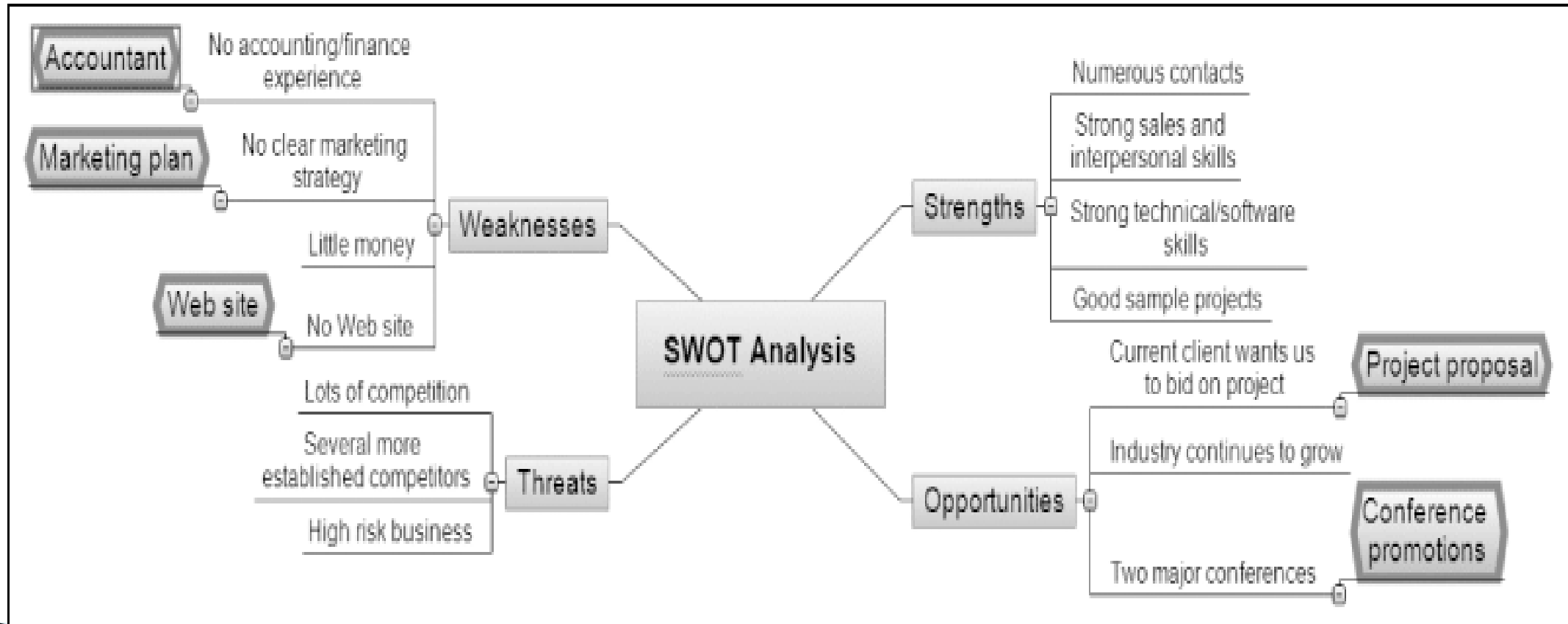
Project Start

Project Finish

Strategic Planning and Project Selection

- ▶ **Strategic planning** involves determining long-term objectives, predicting future trends, and projecting the need for new products and services
- ▶ Organizations often perform a **SWOT analysis**
 - analyzing **S**trengths, **W**eaknesses, **O**pportunities, and **T**hreats
- ▶ As part of strategic planning, organizations
 - identify potential projects
 - use realistic methods to select which projects to work on
 - formalize project initiation by issuing a project charter

Mind Map of a SWOT Analysis to Help Identify Potential Projects



Methods for Selecting Projects

- ▶ There are usually more projects than available time and resources to implement them
- ▶ Methods for selecting projects include:
 - Focusing on broad organizational needs
 - Categorizing information technology projects
 - Performing net present value or other financial analyses
 - Using a weighted scoring model
 - Implementing a balanced scorecard

Focusing on Broad Organizational Needs

- ▶ It is often difficult to provide strong justification for many IT projects, but everyone agrees they have a high value
- ▶ Three important criteria for projects:
 - There is a ***need*** for the project
 - There are ***funds*** available
 - There's a strong ***will*** to make the project succeed

Categorizing IT Projects

- ▶ One categorization is whether the project addresses
 - **a problem** (solve an existing issue that negatively affects)
 - **an opportunity** (take advantage of new technologies, trends, or business chances to improve competitiveness or growth)
 - **a directive** (initiated because of a mandate, regulation, or top-level management order, not necessarily because of a problem or opportunity)
- ▶ Another categorization is how long it will take to do and when it is needed
- ▶ Another is the overall priority of the project

Financial Analysis of Projects

- ▶ Financial considerations are often an important consideration in selecting projects
- ▶ Three primary methods for determining the projected financial value of projects:
 - Net present value (NPV) analysis
 - Return on investment (ROI)
 - Payback analysis

Net Present Value Analysis

- ▶ **Net present value** (NPV) analysis is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time
- ▶ It is a **financial metric** used to evaluate the profitability of an investment, project, or decision by comparing the value of expected future cash inflows to the initial investment cost, after adjusting for the **time value of money**.
- ▶ Projects with a positive NPV should be considered if financial value is a key criterion
- ▶ The higher the NPV, the better

Net Present Value Analysis

Formula:

$$NPV = \sum_{t=1}^n \frac{C_t}{(1+r)^t} - C_0$$

Where:

- C_t = cash inflow (or outflow if negative) at time t
- r = discount rate (required rate of return or cost of capital)
- t = time period (1, 2, 3, ..., n)
- C_0 = initial investment (cash outflow at $t = 0$)

Net Present Value Analysis (Example)

- ▶ Suppose you invest **\$1,000** today in a project that will return **\$400 per year** for 3 years. The discount rate is **10%**.

$$NPV = \frac{400}{(1 + 0.1)^1} + \frac{400}{(1 + 0.1)^2} + \frac{400}{(1 + 0.1)^3} - 1000$$

$$NPV = 363.64 + 330.58 + 300.53 - 1000 = -5.25$$

Net Present Value Analysis (Example)

- ▶ Project R requires an initial investment of \$45,000 and expects to generate \$30,000 annually for two years, with an 8% discount rate. To estimate the NPV, the company that wants to invest in Project R finds the individual NPV of both cash flows and adds the results. Then, it deducts the initial investment from the sum of the NPVs.
- ▶ **NPV of Project R = $[\$30,000 / (1+0.08)^1] + [\$30,000 / (1+0.08)^2] - \$45,000$**
NPV = $(\$27,777.78 + \$25,720.17) - \$45,000 = \$8,497.95$

Return on Investment

- ▶ **Return on investment (ROI)** is a **performance measure** that tells you how much profit or benefit you gain from an investment compared to its cost.
- ▶ It is calculated by subtracting the project costs from the benefits and then dividing by the costs

$$ROI = \frac{\text{Net Benefit (Gain from Investment - Cost of Investment)}}{\text{Cost of Investment}} \times 100\%$$

- ▶ The higher the ROI, the better
- ▶ Many organizations have a **required rate of return** or minimum acceptable rate of return on investment for projects
- ▶ **Internal rate of return (IRR)** can be calculated by finding the discount rate that makes the NPV equal to zero

Return on Investment (Example)

- You invest **\$10,000** in a project.
- The project generates **\$15,000** in benefits (savings or revenue).
- Net benefit = $15,000 - 10,000 = \mathbf{\$5,000}$
- $\text{ROI} = (5000 / 10000) \times 100\% = 50\%$

Payback Analysis

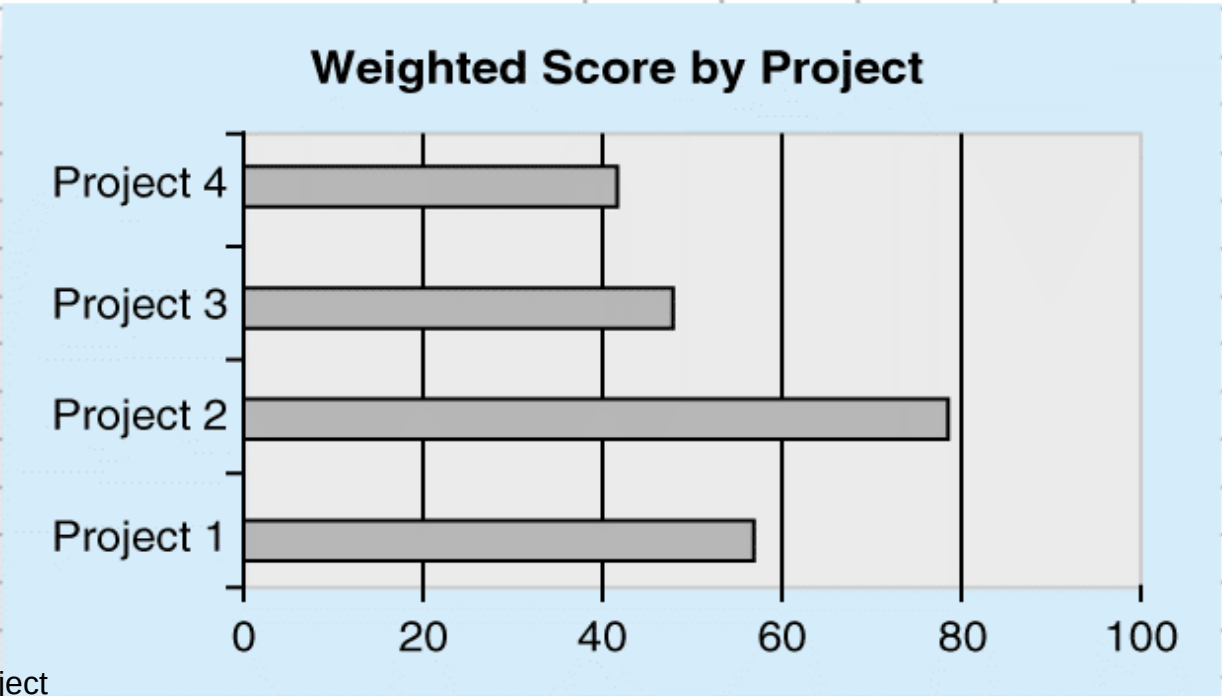
- ▶ Another important financial consideration is payback analysis
 - *“How many years (or months) will it take until I get my money back?”*
- ▶ The **payback period** is the amount of time it will take to recoup, in the form of net cash inflows, the total dollars invested in a project
- ▶ Payback occurs when the net cumulative discounted benefits equal the costs
- ▶ Many organizations want IT projects to have a fairly short payback period

Weighted Scoring Model

- ▶ A weighted scoring model is a tool that provides a systematic process for selecting projects based on many criteria
 - Identify criteria important to the project selection process
 - Assign weights (percentages) to each criterion so they add up to 100%
 - Assign scores to each criterion for each project
 - Multiply the scores by the weights and get the total weighted scores
- ▶ The higher the weighted score, the better

Sample Weighted Scoring Model for Project Selection

	A	B	C	D	E	F
1	Criteria	Weight	Project 1	Project 2	Project 3	Project 4
2	Supports key business objectives	25%	90	90	50	20
3	Has strong internal sponsor	15%	70	90	50	20
4	Has strong customer support	15%	50	90	50	20
5	Uses realistic level of technology	10%	25	90	50	70
6	Can be implemented in one year or less	5%	20	20	50	90
7	Provides positive NPV	20%	50	70	50	50
8	Has low risk in meeting scope, time, and cost goals	10%	20	50	50	90
9	Weighted Project Scores	100%	56	78.5	50	41.5
10						



Implementing a Balanced Scorecard

- ▶ Drs. Robert Kaplan and David Norton developed this approach to help select and manage projects that align with business strategy
- ▶ The **Balanced Scorecard** is a framework that tracks organizational performance across **financial, customer, internal process, and learning/growth** perspectives—ensuring balance between short-term results and long-term sustainability.
- ▶ A framework that measures success across **Financial, Customer, Internal Process, and Learning & Growth** perspectives, instead of looking only at money.

Implementing a Balanced Scorecard

- ▶ Organizations often have many project proposals but **limited resources**.

The Balanced Scorecard helps by checking:

- ☐ Does the project align with the **strategy**?
- ☐ Does it contribute across **all four perspectives** (not just financial)?
- ☐ This ensures that the chosen projects are not only profitable but also support **customer satisfaction, internal efficiency, and long-term growth**.

How **Balanced Scorecard** Guides Project Selection

- **Financial Perspective**

- Does the project increase revenue, reduce costs, or improve ROI?
- Example: A project that automates billing may save costs and improve margins.

- **Customer Perspective**

- Will it improve customer experience, satisfaction, or market share?
- Example: A project introducing faster delivery improves customer satisfaction.

- **Internal Process Perspective**

- Does it streamline operations, reduce defects, or speed up processes?
- Example: A supply chain optimization project reduces delays and waste.

- **Learning & Growth Perspective**

- Does it enhance employee skills, innovation, or organizational capabilities?
- Example: A digital training platform project improves staff knowledge and adaptability.

Developing a Project Charter

- ▶ After deciding what project to work on, it is important to let the rest of the organization know
- ▶ A **project charter** is a document that formally recognizes the existence of a project and provides direction on the project's objectives and management
- ▶ Key project stakeholders should sign a project charter to acknowledge agreement on the need and intent of the project; a signed charter is a key output of project integration management

Inputs for Developing a Project Charter

- ▶ A project statement of work
- ▶ A business case
- ▶ Agreements
- ▶ Enterprise environmental factors
- ▶ **Organizational process assets**, which include formal and informal plans, policies, procedures, guidelines, information systems, financial systems, management systems, lessons learned, and historical information

Project Charter for the DNA-Sequencing Instrument Completion Project

Project Title: DNA-Sequencing Instrument Completion Project

Date of Authorization: February 1

Project Start Date: February 1

Projected Finish Date: November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated \$1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The DNA-sequencing instrument project has been underway for three years. It is a crucial project for our company. This is the first charter for the project, and the objective is to complete the first version of the software for the instrument in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

Project Charter (cont..)

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the DNA sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com

Sign-off: (Signatures of all the above stakeholders)

Ahmed Abrams
Susan Johnson
Erik Haus
Maggie Elliot

Nick Carson
Renyong Chi
Bill Strom

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

"I want to be heavily involved in this project. It is crucial to our company's success, and I expect everyone to help make it succeed." —Ahmed Abrams

"The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me." —Renyong Chi

Developing a Project Management Plan

- ▶ A **project management plan** is a document used to coordinate all project planning documents and help guide a project's execution and control
- ▶ Plans created in the other knowledge areas are subsidiary parts of the overall project management plan

Common Elements of a Project Management Plan

- ▶ Introduction or overview of the project
- ▶ Description of how the project is organized
- ▶ Management and technical processes used on the project
- ▶ Work to be done, schedule, and budget information

Table 4-2. Sample Contents for a Software Project Management Plan (SPMP)

MAJOR SECTION HEADINGS	SECTION TOPICS
Overview	Purpose, scope, and objectives; assumptions and constraints; project deliverables; schedule and budget summary; evolution of the plan
Project Organization	External interfaces; internal structure; roles and Responsibilities
Managerial Process Plan	Start-up plans (estimation, staffing, resource acquisition; and project staff training plans); work plan (work activities, schedule, resource, and budget allocation); control plan; risk management plan; closeout plan
Technical Process Plans	Process model; methods, tools, and techniques; infrastructure Plan; product acceptance plan
Supporting Process Plans	Configuration management plan; verification and validation plan; documentation plan; quality assurance plan; reviews And audits; problem resolution plan; subcontractor management plan; process improvement plan
IEEE Standard 1058-1998.	

Directing and Managing Project Work

- ▶ Involves managing and performing the work described in the project management plan
- ▶ The majority of time and money is usually spent on execution
- ▶ The application area of the project directly affects project execution because the products of the project are produced during execution



Source: Andy Crowe

Coordinating Planning and Execution

- ▶ Project planning and execution are intertwined and inseparable activities
- ▶ Those who will do the work should help to plan the work
- ▶ Project managers must solicit input from the team to develop realistic plans

Providing Leadership and a Supportive Culture

- ▶ Project managers must lead by example to demonstrate the importance of creating and then following good project plans
- ▶ Organizational culture can help project execution by
 - providing guidelines and templates
 - tracking performance based on plans
- ▶ Project managers may still need to break the rules to meet project goals, and senior managers must support those actions

Project Execution Tools and Techniques

- ▶ **Expert judgment:** Experts can help project managers and their teams make many decisions related to project execution
- ▶ **Meetings:** Meetings allow people to develop relationships, pick up on important body language or tone of voice, and have a dialogue to help resolve problems.
- ▶ **Project management information systems:** There are hundreds of project management software products available on the market today, and many organizations are moving toward powerful enterprise project management systems that are accessible via the Internet
- ▶ See the What Went Right? example of Kuala Lumpur's Integrated Transport Information System on p. 169

Monitoring and Controlling Project Work

- ▶ Changes are inevitable on most projects, so it's important to develop and follow a process to monitor and control changes
- ▶ Monitoring project work includes collecting, measuring, and disseminating performance information
- ▶ A **baseline** is the approved project management plan plus approved changes

Performing Integrated Change Control

- ▶ Three main objectives are:
 - Influencing the factors that create changes to ensure that changes are beneficial
 - Determining that a change has occurred
 - Managing actual changes as they occur

Change Control on Information Technology Projects

- ▶ Former view: The project team should strive to do exactly what was planned on time and within budget
- ▶ Problem: Stakeholders rarely agreed up-front on the project scope, and time and cost estimates were inaccurate
- ▶ Modern view: Project management is a process of constant communication and negotiation
- ▶ Solution: Changes are often beneficial, and the project team should plan for them

Change Control System

- ▶ A **change control system** is a formal, documented process that describes when and how official project documents and work may be changed
- ▶ Describes who is authorized to make changes and how to make them

Change Control Board (CCB)

- ▶ A **change control board** is a formal group of people responsible for approving or rejecting changes on a project
- ▶ CCBs provide guidelines for preparing change requests, evaluate change requests, and manage the implementation of approved changes
- ▶ Includes stakeholders from the entire organization

Making Timely Changes

- ▶ Some CCBs only meet occasionally, so it may take too long for changes to occur
- ▶ Some organizations have policies in place for time-sensitive changes
 - “48-hour policy” allows project team members to make decisions, then they have 48 hours to reverse the decision pending senior management approval
 - Delegate changes to the lowest level possible, but keep everyone informed of changes

Configuration Management

- ▶ **Configuration management** ensures that the descriptions of the project's products are correct and complete
- ▶ Involves identifying and controlling the functional and physical design characteristics of products and their support documentation
- ▶ Configuration management specialists identify and document configuration requirements, control changes, record and report changes, and audit the products to verify conformance to requirements
- ▶ See www.icmhq.com for more information

Closing Projects or Phases

- ▶ To close a project or phase, you must finalize all activities and transfer the completed or cancelled work to the appropriate people
- ▶ Main outputs include
 - Final product, service, or result transition
 - Organizational process asset updates

Using Software to Assist in Project Integration Management

- ▶ Several types of software can be used to assist in project integration management
 - Documents can be created with word processing software
 - Presentations are created with presentation software
 - Tracking can be done with spreadsheets or databases
 - Communication software can facilitate communications
 - Project management software can pull everything together and show detailed and summarized information

Sample Portfolio Management Software Screen



Source: www.projectmanager.com

Chapter Summary

- ▶ Project integration management involves coordinating all of the other knowledge areas throughout a project's life cycle
- ▶ Main processes include
 - Develop the project charter
 - Develop the project management plan
 - Direct and manage project execution
 - Monitor and control project work
 - Perform integrated change control
 - Close the project or phase